Table 5
Published flow reductions from paired catchment analyses, after Scott et al. (2000), Hickel (2001), Nandakumar and Mein (1993) and Fahey and Jackson (1997) compared to estimated reductions in this study

Catchment	Year	Rainfall[a] (mm)	Total flow reduction (%)	Low flow reduction (%)	Estimated from Fig. 4	
					Total flow reduction (%)[b]	Low flow reduction (%)[c]
Cathedral Peak 2	21	1516 (+80)	50	48	57	54
Cathedral Peak 3	18	1556 (+52)	60	53	62[d]	53
Lambrechtsbos A	13	1111 (−23)	41	34	41	38
Lambrechtsbos B	18	1079 (−9)	69	78	58	63
Biesievlei	20	1388 (+6)	52	62	52	60
Redhill	9	783 (−93)	66	100	75	100
Stewarts Ck 5	20	1249 (+93)	69		64	
Glendhu	Average reduction		27	<20	32	35

[a] Rainfall refers to the rainfall in the year used for comparison of results. The value in brackets refers to the deviation from the mean annual rainfall for the period of record.
[b] Total flow reduction calculated by $\sum Y / \sum (a+Y)$ for all deciles.
[c] Low flow reduction calculated by $\sum Y / \sum (a+Y)$ for 70, 80, 90 and 100th percentiles.
[d] For Cathedral Peak 3 the a and Y values for the 10 and 20th percentiles were excluded as the values of a were lower then the values of the 30–100th percentiles.

catchments and the lower deciles at Lambrechtsbos B. The T_{half} from Glendhu 2 appears to be substantially lower than other published data (Fahey and Jackson, 1997).

4.4. Comparison with paired catchment studies

A further check on the overall model performance is a comparison with published results of paired catchment studies. The data that can be compared with our results are presented in Table 5 and can be broadly compared with Fig. 4. These data are reductions in years with near average annual rainfall, and at a time after treatment when maximum changes in streamflow have occurred. Table 5 also includes estimates on the total and low flow reductions calculated from this study. Results from Pine Creek and Traralgon Creek are not included in Table 5 as these catchments are not paired. Exact comparisons are impossible because of the rainfall variability, and lack of calibration period for Redhill. Despite this, Table 5 shows that total and low flow reductions estimated from our study are comparable to the results from paired catchment studies, indicating that our simple model has successfully removed the rainfall signal.

4.5. Zero flow days

As this analysis could only be applied, where there was consistent drying up of streams, it was confined to Stewarts Creek, Pine Creek and Redhill catchments. The model returned values of E of 0.95, 0.99 and 0.99, respectively. The t-tests on b and ΔN_{zero} returned significant results at the 0.05 level for both parameters at all three catchments. The climate adjusted zero flow days are shown in Fig. 5. The increases in zero flow days are substantial with flows confined to less than 50% of the time by year 8 at Stewarts Ck and Pine Ck and year 11 at Redhill. The latter has changed from an almost permanent to a highly intermittent stream. The curves are also in sensible agreement with the flow reductions in Fig. 4.

5. Discussion

The aims of the project have largely been met. The general characterisation of FDCs and adjustment for climate has been very encouraging given the task of fitting our model to 10 flow percentiles, for 10 different catchments (resulting in 100 model fits) with